REVOLUTION RACING

Design & Engineering
Portfolio

T25

Città di Giaveno

FAMAR

PA SC AL

1 - RESEARCH

Summary

Cover	1
1 - Research	2
2 - Design concepts	3
3 - 3D modelling	4
4 - CFD simulations	5
5 - Car design	6
Drawings	10
6 - Crafting & testing	11

1.1 Car development goals

Performance
(fastest race times possible)

Compliance
(meet all race regulations)

Durability
(solid structure, withstanding all race launches)

1.2 Car Dynamics

1.2.1 Rigid body dynamics

We began by studying rigid body dynamics, treating the car as a rigid body subject to forces and torques. For this project we adopted the following simplifying assumptions:

The center of mass keeps a constant elevation, allowing a 2D analysis.

We consider only wheel friction, neglecting tether-guide friction as it is minimal and hard to measure.

The car's rotation is limited to yaw, as roll and pitch are constrained by the track.

The car is symmetric and the canister thrust is aligned with the center of mass in the XY plane.

1.2.2 Aerodynamic forces

$$d_f = \frac{1}{2}\rho C_d A v^2$$ Drag, acting in the direction of motion

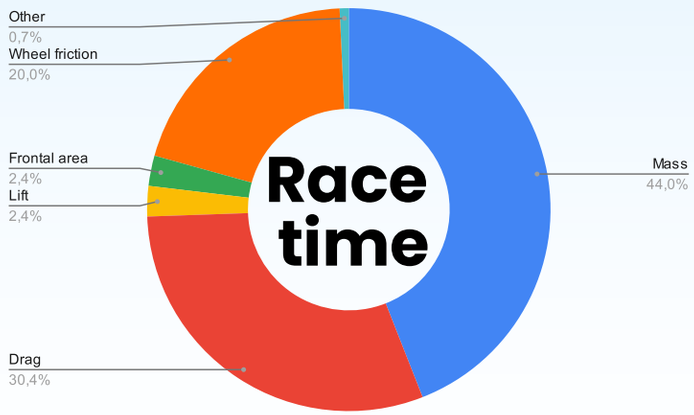
$$d_l = \frac{1}{2}\rho C_t A v^2$$ Lift, acting in the Z direction

$$\tau = \frac{1}{2}\rho V^2 A L C_M \alpha$$ Aerodynamic torque

As these formulas show we should aim at minimizing the frontal area of the model.

1.3 Race model

Based on the dynamics study, we implemented a time-stepping simulation in C++. Given a set of vehicle parameters, the simulator computes race time. The model was developed to quantify the impact of individual factors on lap time and to guide design decisions during the development phase.

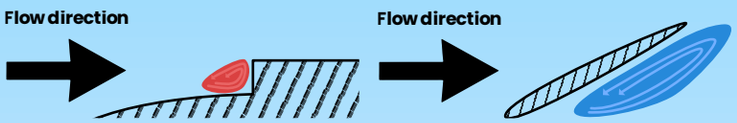


2 - DESIGN CONCEPTS

2.1 Aerodynamics

After understanding the car's basic behavior, we analyzed the underlying physics and aerodynamics to guide the design.

2.1.1 Recirculation



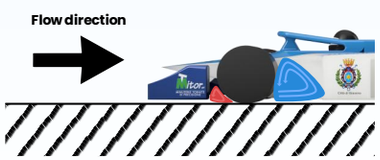
It causes:

Energy loss
(caused by the interaction of the recirculation region and the laminar flow)

Turbulence
(that increases drag)

We can avoid it by employing tunnels to recycle turbulent air in clean flow, or by blockage with car material.

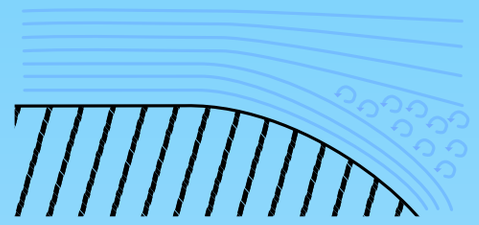
2.1.2 Wheel aerodynamics



Increased energy loss, turbulence, and drag because of different pressure zones.

•Design aim: deflect airflow away from wheels to minimize these losses.

2.1.3 Coanda effect



The Coanda effect—air adhering to smooth surfaces—can guide flow beneficially or, if ignored, cause adverse effects.

2.1.4 Vortices and endplates

Vortices should be avoided as they cause turbulence and drag. Often formed at sharp wing tips, endplates are employed to block them

2.2 Vehicle dynamics

2.2.1 Inertia

Mass reduction preserves impulse and increases speed. Rotational inertia of the car affects yaw stability

2.2.2 Tipping moment

Occurs when the canister's thrust is offset from the vehicle's center of mass, creating torque that wastes part of the thrust.

2.2.3 Normal force

Thanks to stronger lift normal force can be reduced thus reducing rolling resistance of the wheels. Lift should not become so great that the car destabilizes

2.3 Design goals

Minimize mass as much as possible

Minimize aerodynamic drag

Minimize tipping moment

Keep positive lift approximately to 15/20% of the weight

2.4 Design approach

2.4.1 Component separation

Divided in: front wing, rear wing, and body. This allowed to split work, isolate performance variables, and reduce simulation time.

2.4.2 Iterative design

New designs were evaluated and refined against the previous versions. Single component optimizations preferred to whole car ones

2.4.3 CFD Design evaluation

We tested all components separately in CFD (testing approach for the components is explained in sections 5.1.1, 5.2.1 and 5.3.1). Then we took the 2 best front wings, rear wings and bodies and combined them in 8 full designs from which we took the best one

3 – 3D MODELLING

3.1 CAD Software

We used Fusion 360 to model the entire car, following the organizers' recommendation. It enabled assembly and drawing management, mass estimation, and export to multiple formats. Experienced members trained newcomers to fully leverage its capabilities.

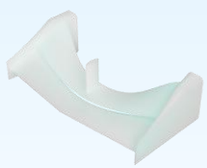
Fusion 360

3.2 Workflow

3.2.1 Component separation and integration

We modelled the car by treating major parts as separate components:

Body
(main structure including the nose cone)

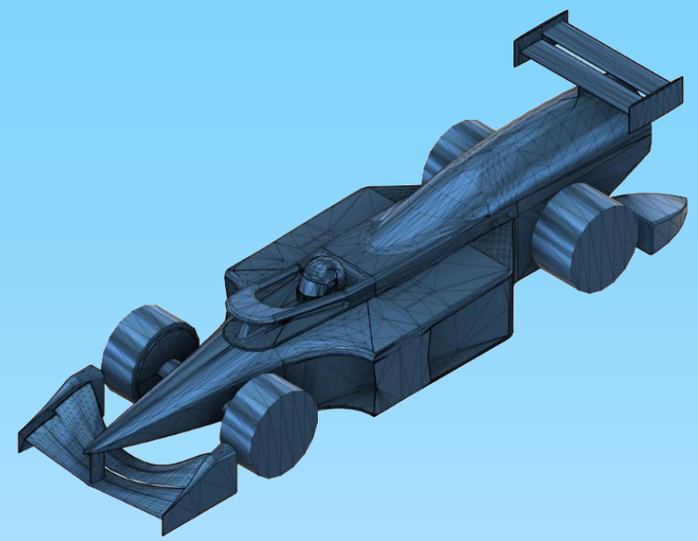


Front wing
(wing and wing support only)

Rear wing
(wing and wing support)

This enabled smoother transitions between parts and avoided forcing a standardized interface. We later joined the bodies and split them into regulation-compliant components

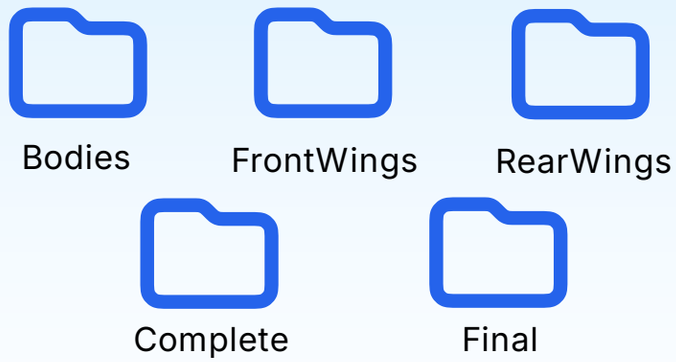
3.2.2 Preparation for simulation



Before CFD analysis we simplified and prepared geometry: we joined bodies (including wheels), removed internal holes, and exported the prepared models as .obj files for meshing.

3.2.3 File structure and organization

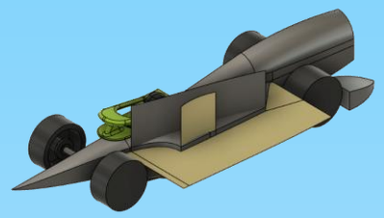
We created a new project and a top-level folder Race1 for the first race. Inside it we kept five main folders:



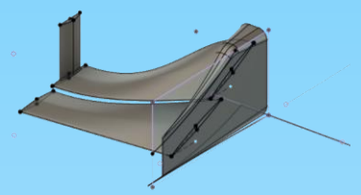
Single components were labeled alphabetically in creation order, while Complete and Final models used three-letter codes indicating body, front wing, and rear wing

3.3 Modelling techniques

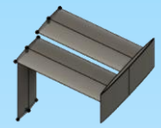
We selected different modelling techniques to suit each component:



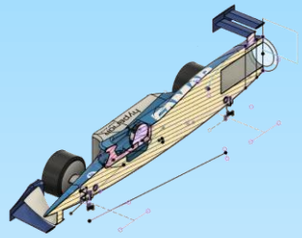
Body
(surface modelling)



Front wing
(advanced solid modelling)

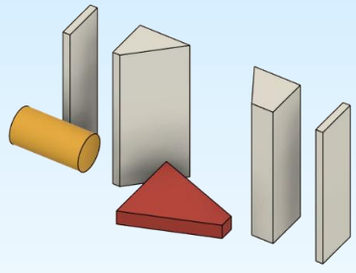


Rear wing
(simpler solid modelling)



Final separation
(solid modelling, body split tools)

3.3.1 Modelling regulations



3.3.2 Tolerances

A minimum tolerance of 0.2 mm was applied throughout to ensure parts remained compliant after manufacturing

3.3.3 Manufacturing adaptation

To facilitate manufacturing we avoided overly thin walls and sharp edges, applying fillets of at least 0.2 mm. Close collaboration with our 3D-printing sponsor FAMAR helped us produce an easily manufacturable design.

4 – CFD SIMULATIONS

4 CFD Simulations

CFD is one of the most important tools for design evaluation and improvement. It allows us to assess a design's performance and to understand airflow, making iterative development faster and more effective. We invested significant effort to ensure our simulations were accurate and numerically robust.

4.1 Software

OpenFOAM

We used the open-source suite OpenFOAM, an advanced, research-grade, Linux-based collection of solvers. We chose OpenFOAM over Ansys for four main reasons:

Ansys student limitations
(restricts mesh size and limits HPC massively)

Flexibility
(highly customizable, can be adapted to wide range of problems)

Transparency
(exposes solver algorithms and parameters, Ansys internal methods are less clear)

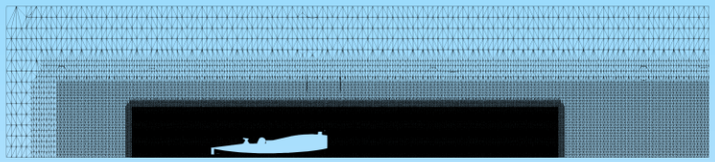
Validated accuracy
(exposes solver algorithms and parameters, Ansys internal methods are less clear)

4.2 Simulations setup

4.2.1 CFD Solver

We used a transient LES solver because the flow around a body of this complexity is inherently turbulent and unsteady. An unsteady solver is therefore the most appropriate choice

4.2.2 Meshing



The mesh was generated with SnappyHexMesh, placing the car inside an enclosure of 1,4 m (length)* 0,3 m (height) *0,6 m (width). It was configured to produce approximately 10 million cells, ensuring high mesh quality near the car while covering large computational domain.

4.2.3 Visualization and postprocessing

ParaView

We used ParaView for visualization and postprocessing. Force values (drag and lift) were computed from the simulation field results using ParaView's tools.

4.2.4 Helper scripts

We developed helper scripts to automate setup, execution, postprocessing, case prep, job submission, ParaView postprocessing, and upload to the team's Drive.

4.3 Simulations validation

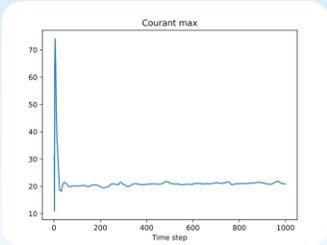
4.3.1 Wind tunnel measurements

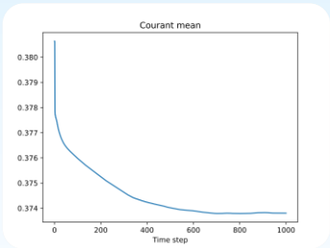
Last year we ran 14 m/s wind-tunnel tests on our previous car, used them to validate OpenFOAM and compare with Ansys Discovery. We averaged the data into one reference, then ran OpenFOAM and Ansys in Explore and Refine modes. Results below.

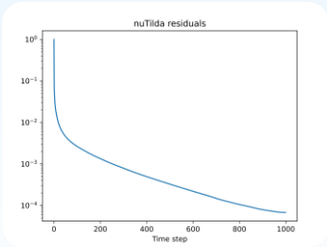
	Drag error	Lift error
OpenFOAM base setup	3.35%	14.87%
Ansys discovery explore mode	64%	-32%
Ansys discovery refine mode	87.30%	-55.15%

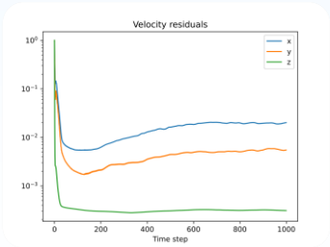
4.3.2 Numerical stability

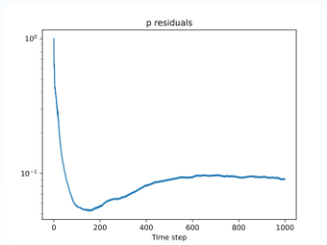
After promising agreement with experiments, we verified numerical stability by checking the Courant's number (max/mean) and checking the convergence of residuals









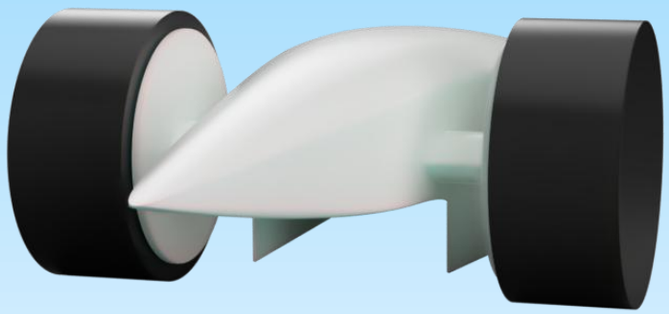


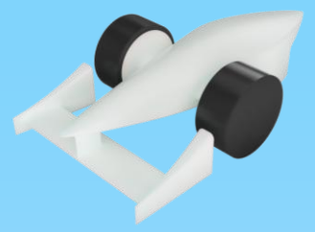
5 – CAR DESIGN

5.1 Front wing design

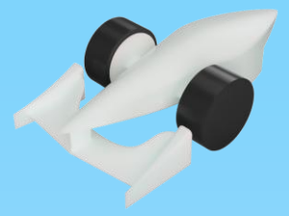
5.1.1 CFD testing

To test only the front wings in CFD, we designed a small component that replicates the real flow on the component and includes the front wheels, which are essential for the wing's aerodynamics.

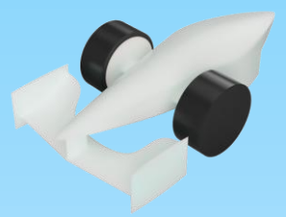




Prototype A
First basic design: flat wing with triangular endplates to deflect air from the wheels.

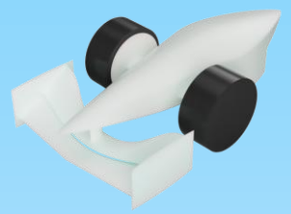


Prototype B
Similar to the first, curved wing deflecting more air from the wheel (refer to 2.1.2).

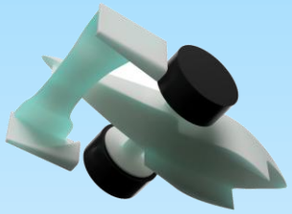


Prototype C
Very similar to B but with flat endplates instead of triangular ones.

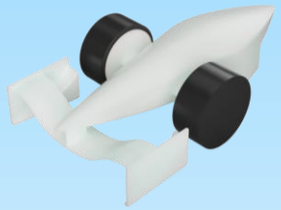
Prototype D
Multi-vane wing with triangular endplates, designed to reduce wake (refer to 2.1.1).



Prototype E
Very similar to D, with flat endplates



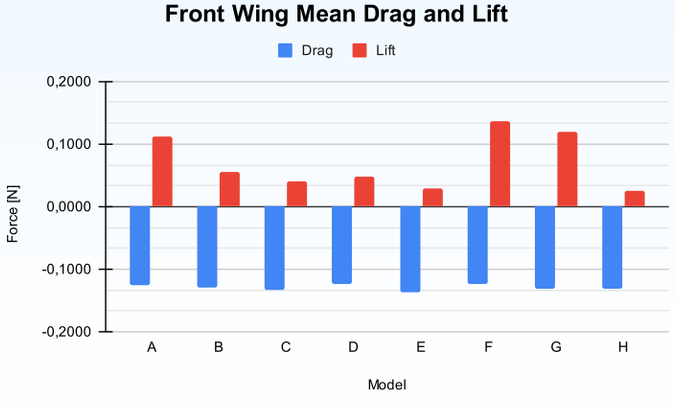
Prototype F
A very different new design, putting the endplates underneath the wing.



Prototype G
Similar to F but with a constant chord wing

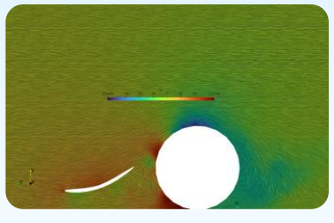
5.1.2 Iterative design

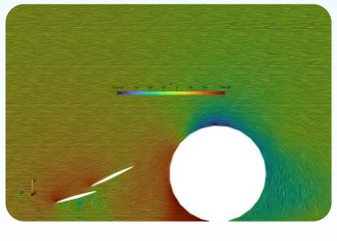
The front wing undergone a total of 7 different design iterations, labeled with the letters A through G, here are the drag and lift results for all the models:



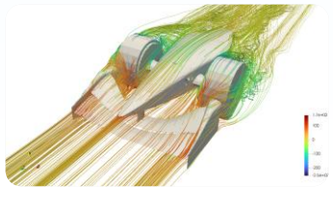
5.1.3 D

This was the overall best wing model, its slitted design allowed it to maintain a smaller wake region behind the wing, by making normal pressure air to get through the opening. Its curved shape also allowed it to deflect air over the wheel.





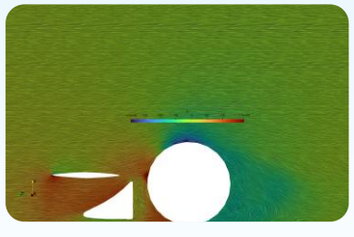
Here is the direct comparison between model D (down) and model B (up), as it can be clearly seen the slitted design reduces the wake region. (refer to 2.1.1 for recirculation and wake)



Streamlines colored by pressure, model D

5.1.4 F

This was the second best design. The main idea in this design was to make the wing higher, letting it to better keep the air over the wheel, while putting the triangular end plates under the wing to still have their deflection effect.



Slice passing through wheel and endplate colored by pressure, model F.

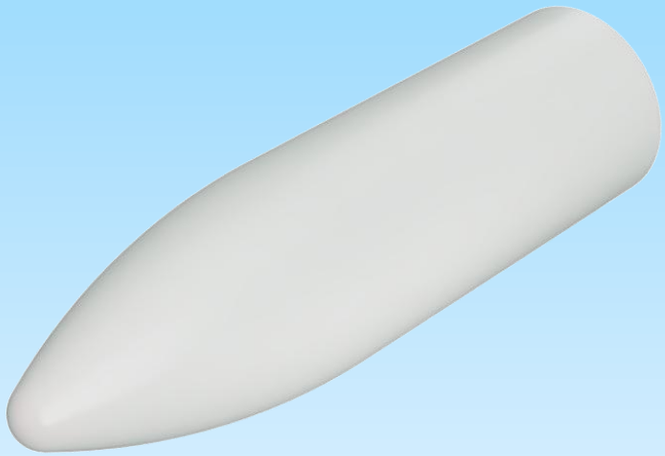
Streamlines colored by pressure, model F

5 – CAR DESIGN

5.2 Rear wing design

5.2.1 CFD testing

Similarly to the front wing we designed once again a component to emulate the flow near the canister vane.



Prototype A
First design, heavily inspired by many word-level car wings, that use this curvy shape.

Prototype B
Another very simple design, featuring just a straight wing and no endplates.

Prototype C
Same as B, plus flat endplates to suppress wing-tip vortex formation (see 2.1.4).

Prototype D
First multi-vane rear wing; straight wings and flat endplates to reduce frontal area (see 1.2.2).

Prototype E
Very similar to D, with flat endplates.

5.2.3 D

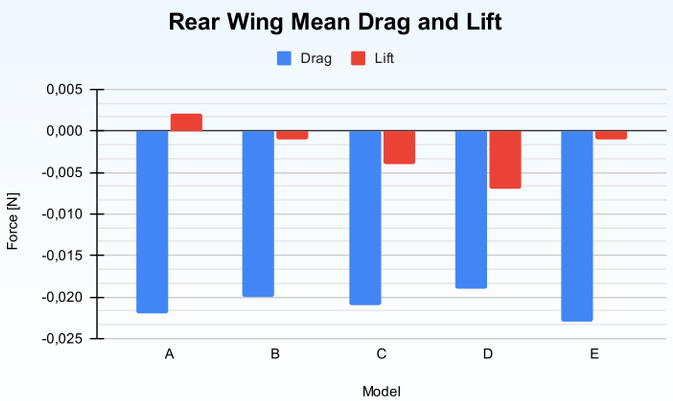
This was the best performing rear wing, the idea in this design was to lower the frontal area (one of the prevalent factors in drag) of the component, this was possible by making really thin single wings while keeping the total thickness of the full wing over the regulatory limit (in fact the regulations only limit total thickness of the multi-vane wing and not the thickness of the single sub wings).

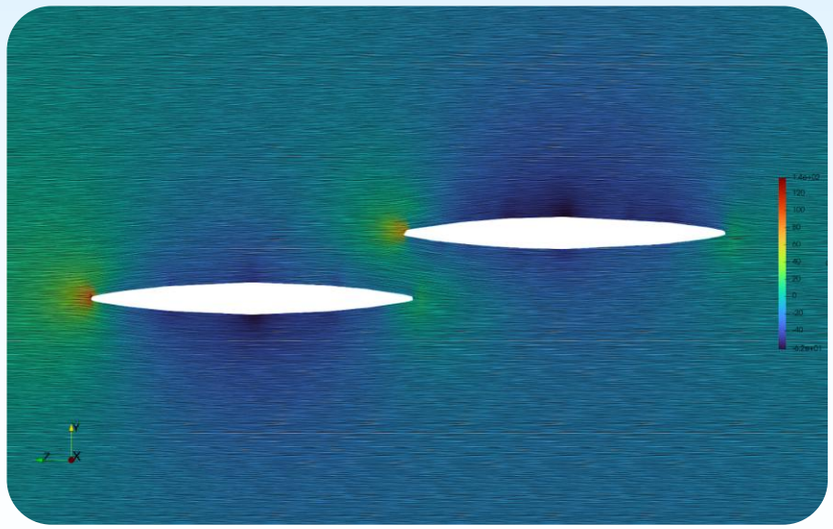
5.2.4 B

This was the second best performing design, it was probably just because of its simplicity, often not overcomplicating things can get good results.

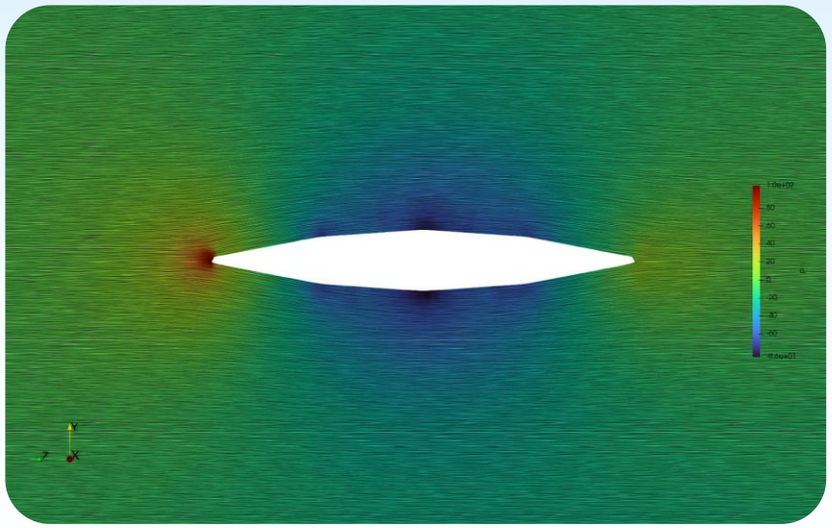
5.2.2 Iterative design

For the front wing we designed 5 different design iterations, labeled by the letters A through E.





Slice passing through wing colored by pressure, model D



Slice passing through wing colored by pressure, model B

5 – CAR DESIGN

5.3 Body design

5.3.1 CFD testing

To test the bodies we decided upon a “standard” front wing and rear wing; these were front wing A and rear wing A. We chose them because they both had average performance and because they were the first models made. So when we had a model to test we would combine the body with front wing A and rear wing A (creating all the models named <body>AA) then we would compare these results and get the best 2 bodies.

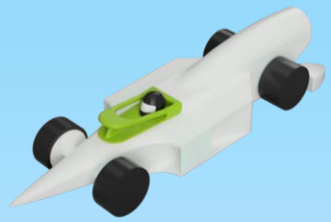
5.3.2 Iterative design

For the body we designed 9 different models, as it is one of the important parts in car aerodynamics.

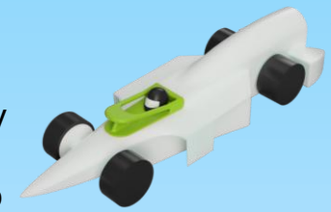
Prototype A
Smoothed triangular body with underfloor tunnel and rear backpods to reduce drag and wake.

Prototype B
Similar to A with a pointier nose, sharper side-pod corners, and a straighter, less curvy profile

Prototype C
Derived from B, side-pod tunnel, side/top separators to isolate clean flow from wheel wake.

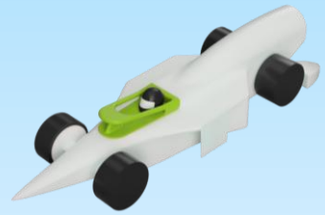


Prototype D
Like C, with a bottom separator fully closing the tunnel to keep underbody airflow clean.

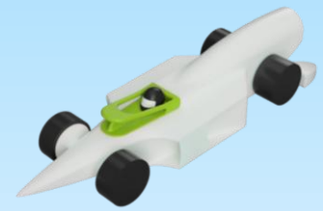


Prototype E
Very similar to C but with bigger backpods and a redesigned rear section.

Prototype F
Basically a copy of C, the only difference was the slight push back of the side separator.

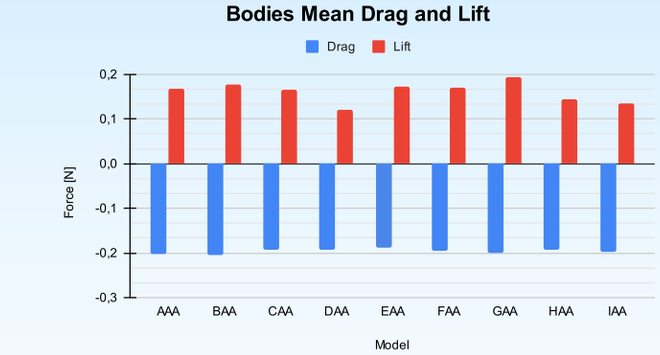


Prototype G
Like F, side separator moved further back, producing a very pointed top-view body.



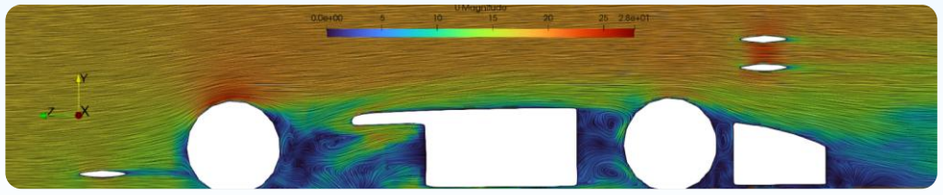
Prototype H
Combination of D and F, basically F with the addition of a bottom separator.

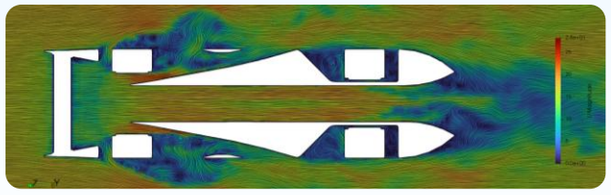
Prototype I
This model is a combination of D and G, again making G with the bottom separator..

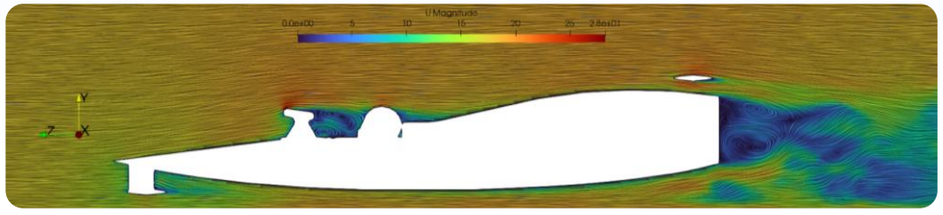


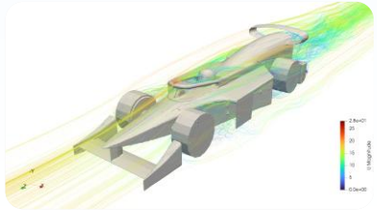
5.3.3 E

The larger backpods of this design improved incredibly the performance compared to C and made it the best design. Also an issue with this design is that the very big backpods could increase drastically the weight of the car.





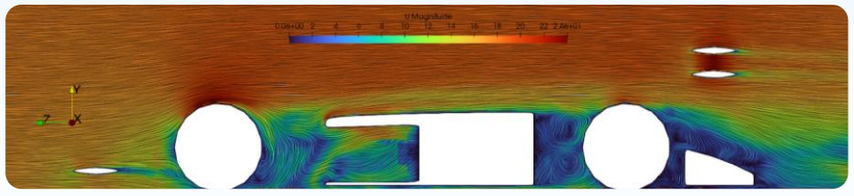


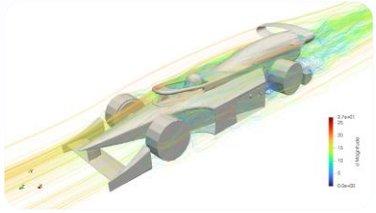


Model EAA

5.3.4 D

This was the second best body design, thanks to the closed side tunnel design creating very clean flow. It also serves the purpose of redirecting air away from the rear wheels.





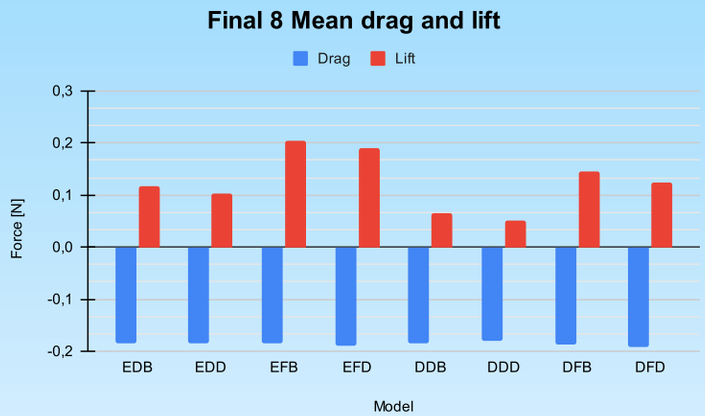
Model DAA

5 – CAR DESIGN

5.4 Final design

5.4.1 Final 8

Starting from the best 2 bodies, best 2 front wings and best 2 rear wings we combined them all creating the final 8 models. These were all simulated and the best model was chosen (as explained in 2.4.3).

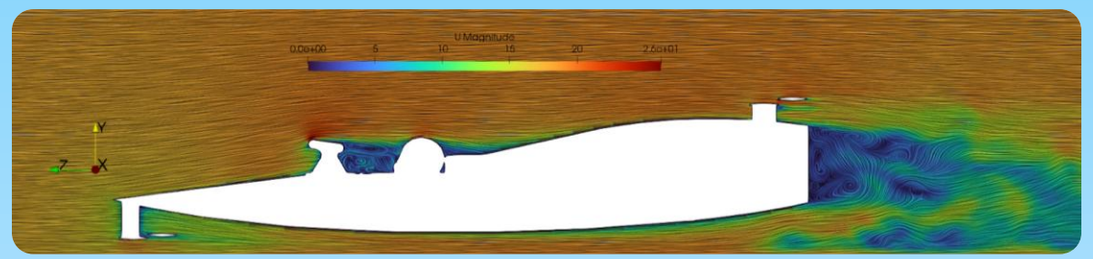


5.4.3 Component separation and final features

Following the design choice, we executed the final modeling phase, splitting the body into three compliant components with custom attachments. We also integrated tether line guides and designed the car's livery.

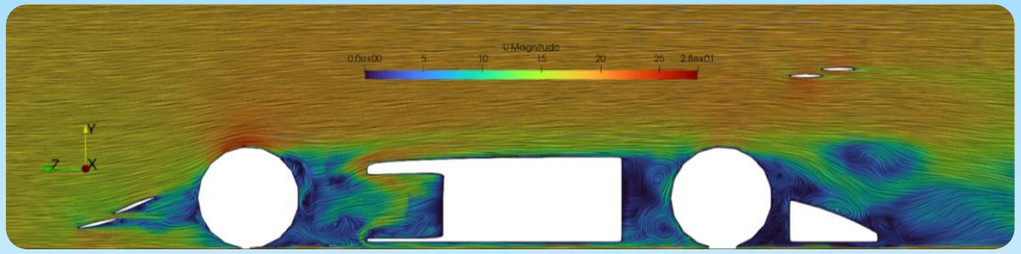
5.4.2 Car DDD

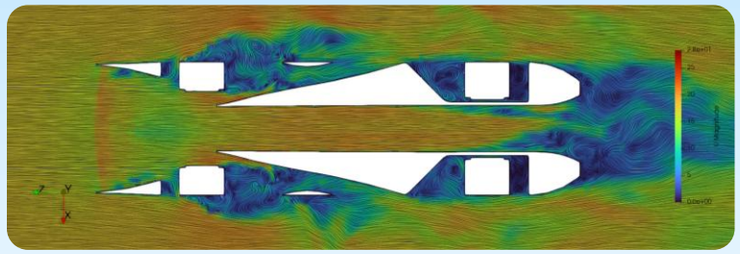
This was chosen as the final design as it performed best in the final round of simulation, this model was composed of the best front wing and best rear wing (D and D), but strangely only from the second best body (D). This proved that this last simulation step was a necessary and useful step, as the best car was not only formed by the best components but also by the best harmony between them



Slice colored by velocity, model DDD. In this image it is shown how the air adheres to the smooth form of the car. (refer to 2.1.3 for Coanda effect)

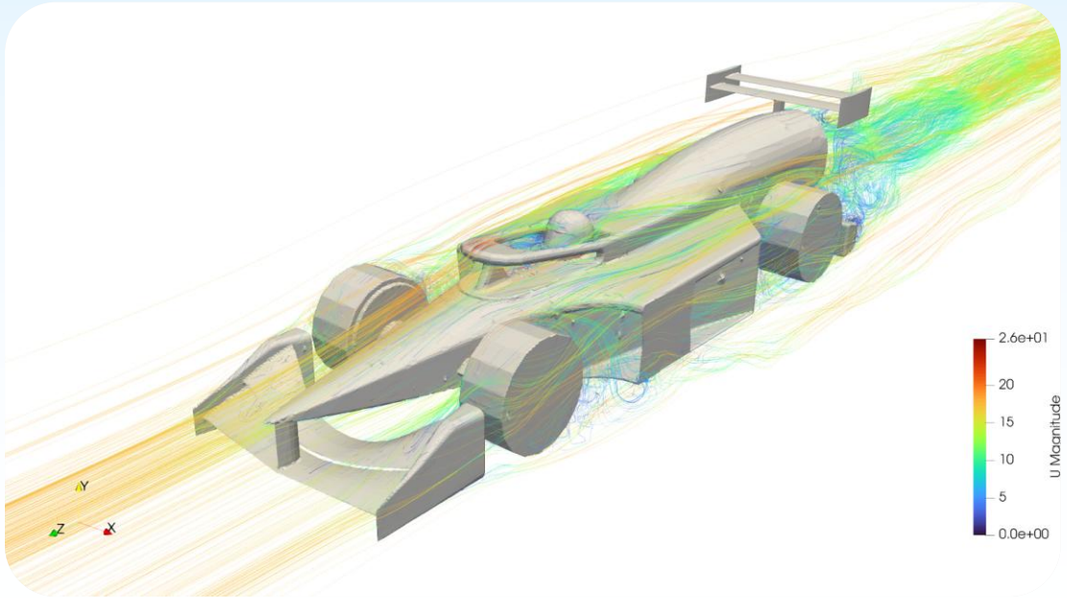
Slice colored by velocity, model DDD

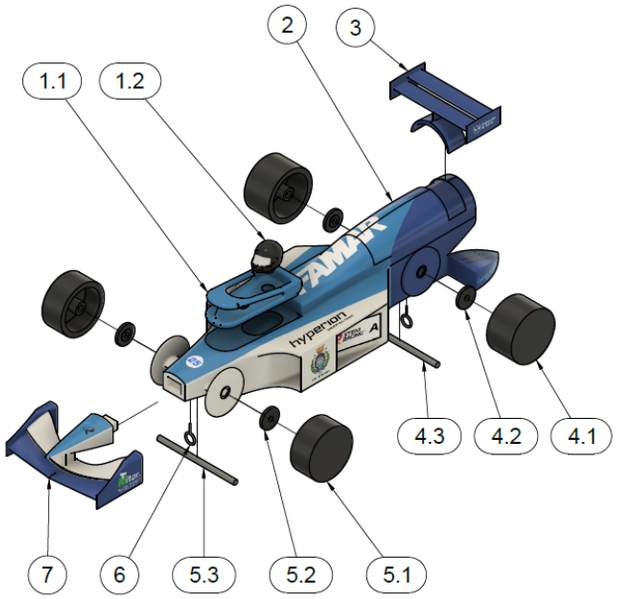




Slice colored by velocity, model DDD. In this image it can be seen how the wake air coming from the front wheel gets funneled in the side pod tunnel and becomes cleaner flow. (refer to 2.1.1 for recirculation and wake)

Streamlines colored by velocity, model DDD

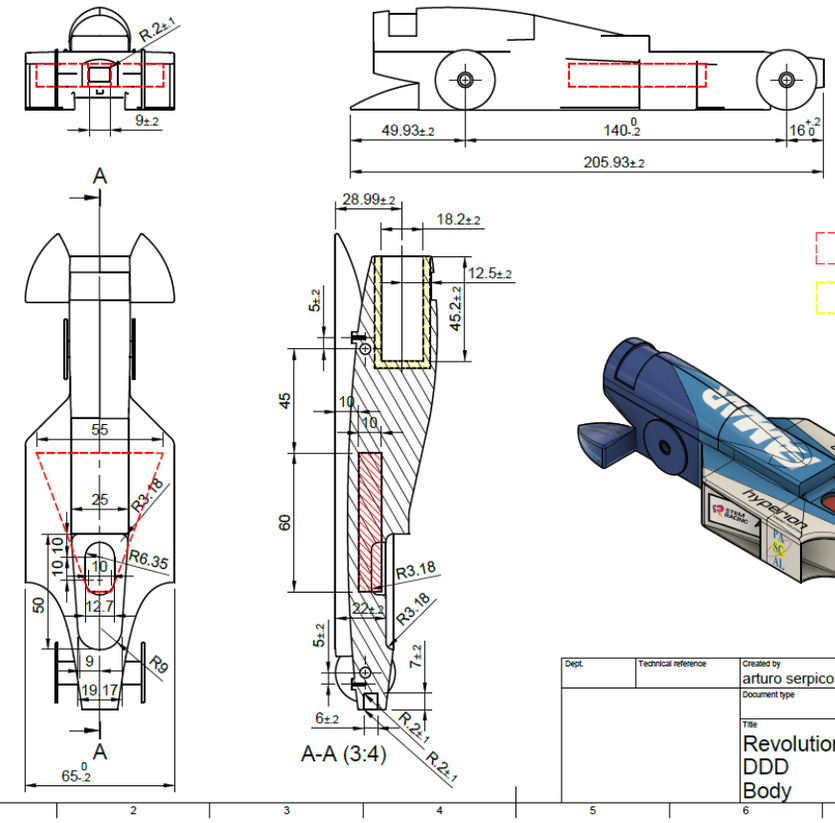




Components List			
Number	Qty.	Name	Material
1	1	Halo assembly	PLA
1.1	1	Halo	PLA
1.2	1	Helmet	PLA
2	1	Body	PLA
3	1	Rear wing	PLA
4	1	Rear wheel assembly	Multiple
4.1	2	Rear wheel	Plastic
4.2	2	Wheel support	Plastic
4.3	1	Wheel axel	Aluminum
5	1	Front wheel assembly	Multiple
5.1	2	Front wheel	Plastic
5.2	2	Wheel support	Plastic
5.3	1	Wheel axel	Aluminum
6	2	Theter line guide	Aluminum
7	1	Front wing	PLA

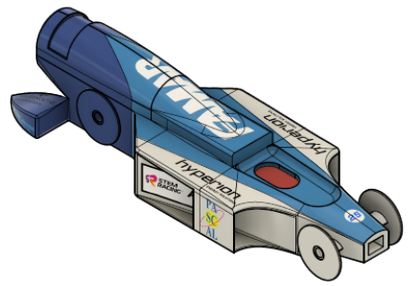
Dept.	Technical reference	Created by	Approved by
		arturo serpico 1/24/2026	
		Document type	Document status
		Title: Revolution Racing DDD Assembly	DWG No.
		Rev. / Date of issue	Sheet 1/9

3:4



Virtual Cargo

Canister safety zone



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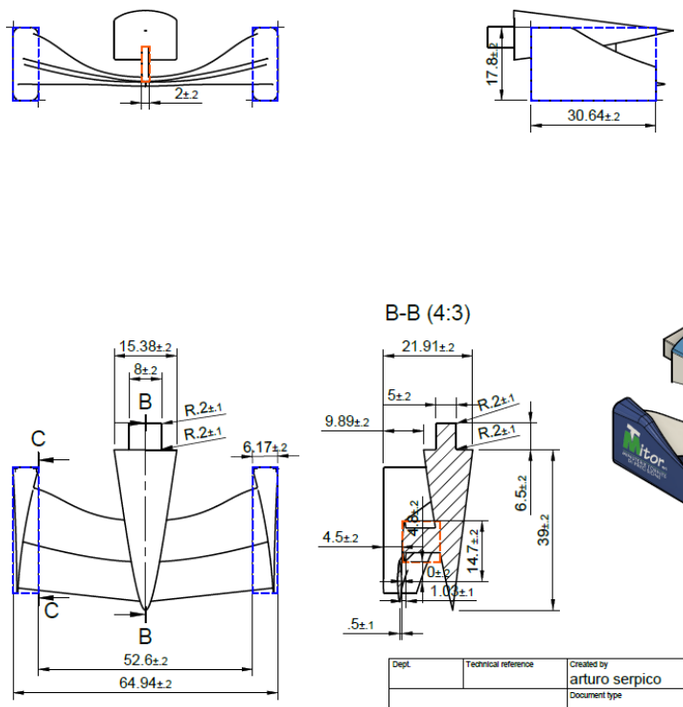
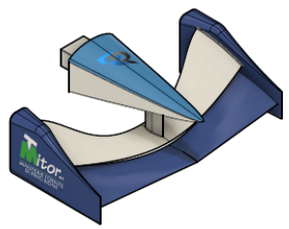
		arturo serpico 1/24/2026	
		Document type	Document status
		Title: Revolution Racing DDD Body	DWG No.
		Rev. / Date of issue	Sheet 2/9

4:3



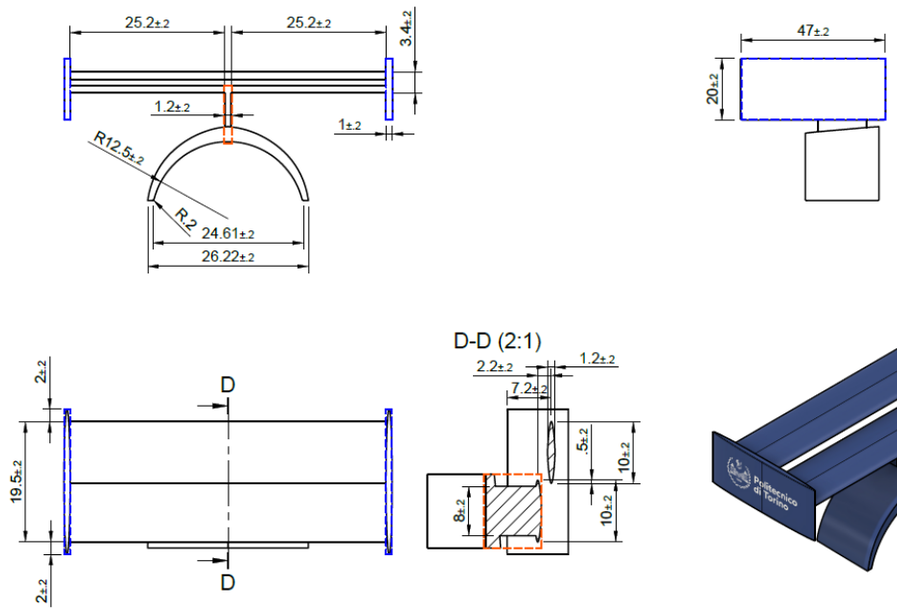


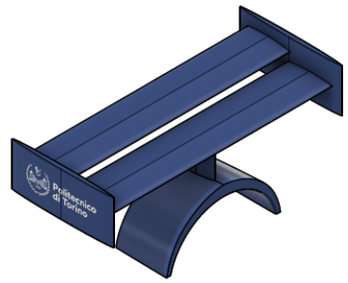
Wing support structure

End plates

Dept.	Technical reference	Created by	Approved by
		arturo serpico 1/24/2026	
		Document type	Document status
		Title: Revolution Racing DDD Front wing	DWG No.
		Rev. / Date of issue	Sheet 3/9

2:1





Wing support structure

End plates

Dept.	Technical reference	Created by	Approved by
		arturo serpico 1/24/2026	
		Document type	Document status
		Title: Revolution Racing DDD Body	DWG No. 10
		Rev. / Date of issue	Sheet 4/9

6 – CRAFTING & TESTING

6.1 Virtual testing

Before starting to manufacture our cars, we subjected our most-performant prototype, DDD, to a complete rule compliance check.

6.2 3D printing

6.2.1 Used 3D printer

Thanks to our main sponsor, FAMAR, we were able to access an industrial 3D printer, the Epsilon W50, that guaranteed us to get high quality and defined models

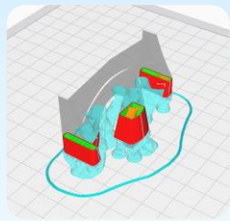
6.2.2 Used material

= chosen

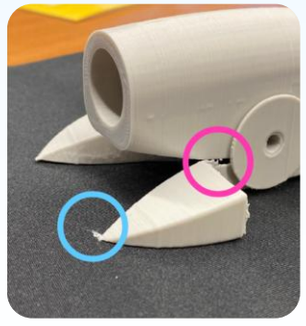
Material	Advantages	Disadvantages
ABS	Light, good mechanical properties	Expands during the printing stage, toxic inalations (safety compliance)
PLA (brand: Filoalfa)	Requires a lower temperature to be deformed, suits with most of the 3D printers, accurate dimensions and detailed printings	Lower mechanical properties
PETG	Good mechanical properties	Requires a high temperature to be deformed, oozing

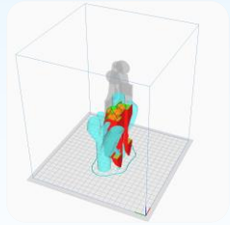
6.2.3 Prototypes

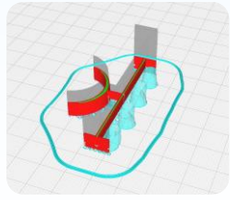
	Prototype 1	Prototype 2	Prototype 3
Material	PLA	PLA	PLA
Layer height	0.2	0.2	0.2
Number of perimeters	2	2	2
Infill	Gyroid, 5% for components, 20% for halo and helmet	Gyroid, 5%	Gyroid, 5%
Orientation	Horizontal	Vertical	Vertical
Type of support	Normal	Tree	Tree
Main problems	- Weight (87 g) - Poor quality (due to supports type and horizontal print)	- Fragility - Poor quality (manufacturing considerations weren't done)	//
Solved problems	//	- Weight (82 g)	- Weight (82 grams) - Model quality



Prototype 1







Prototype 2

Prototype 3

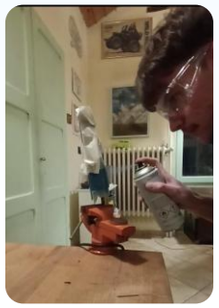
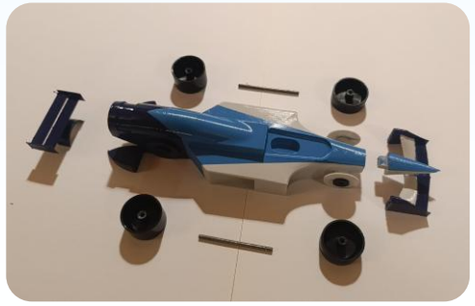
6.3 Physical testing

Once we found a satisfying combination of 3D printing settings, we were able to accomplish tests required by the technical regulations. In particular, tests specified by the following articles were done:

T3.6 (weight), T4.4.5 (halo), T7.13 (wheels)

6.4 Final refinements and safety assessment

On the definitive 3D printed models, we had to do some last work:

- Pass sandpaper on the car's surfaces, to smooth them
- Modify some of STEM Racing's official components – wheels, wheel supports and axles (all shortened) – to fit the car's CAD model
- Apply 2 coats of paint and stickers to match the livery
- Glue wings, wheel supports and tether line guides.

The riskiest stage consisted of the painting one. To protect himself our manufacturing engineer put on a pair of protective glasses and used a support for the car, avoiding skin and eyes damage